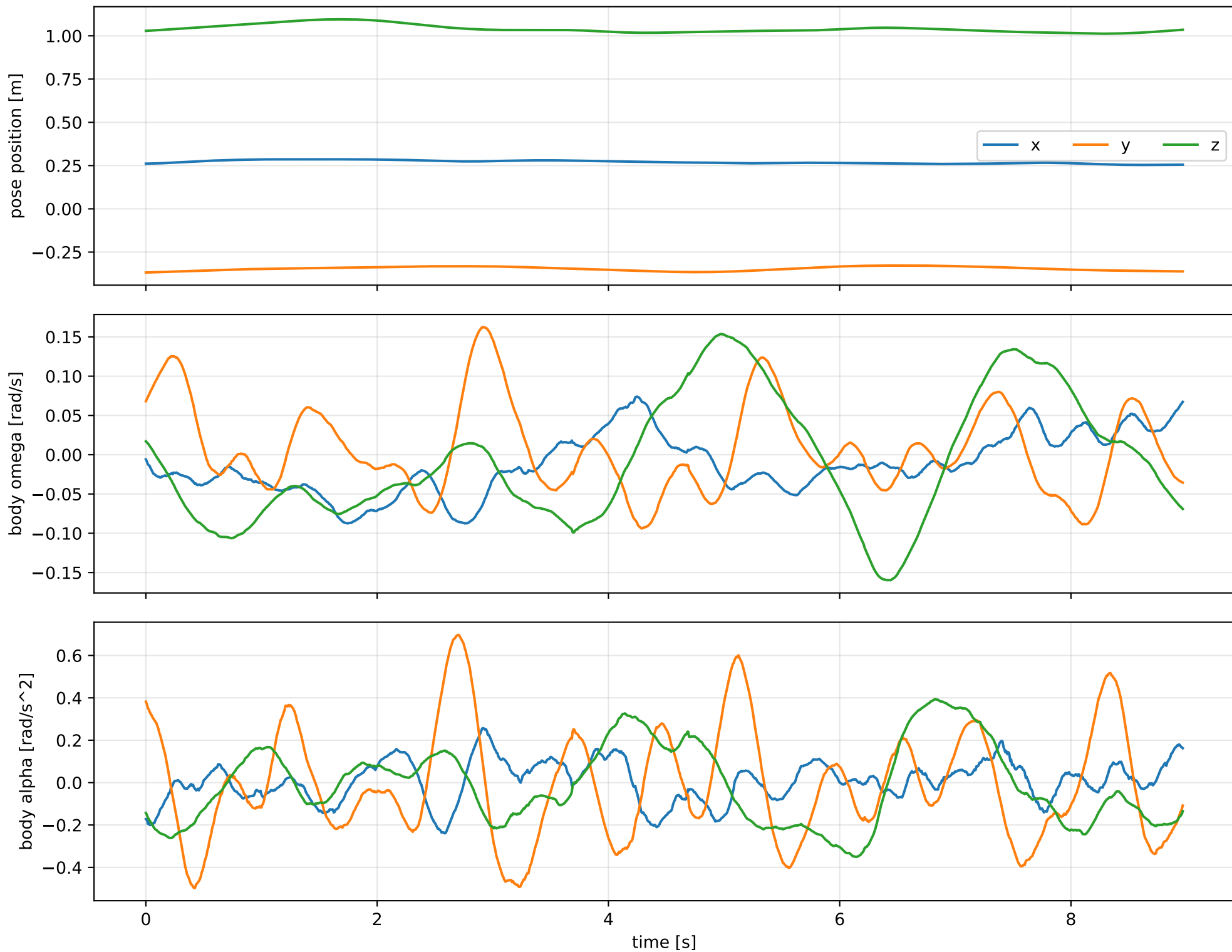
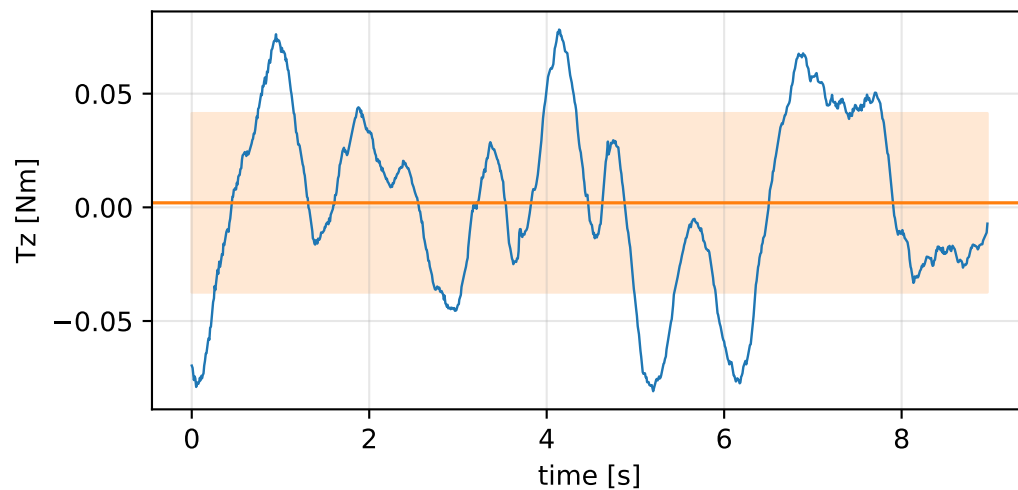
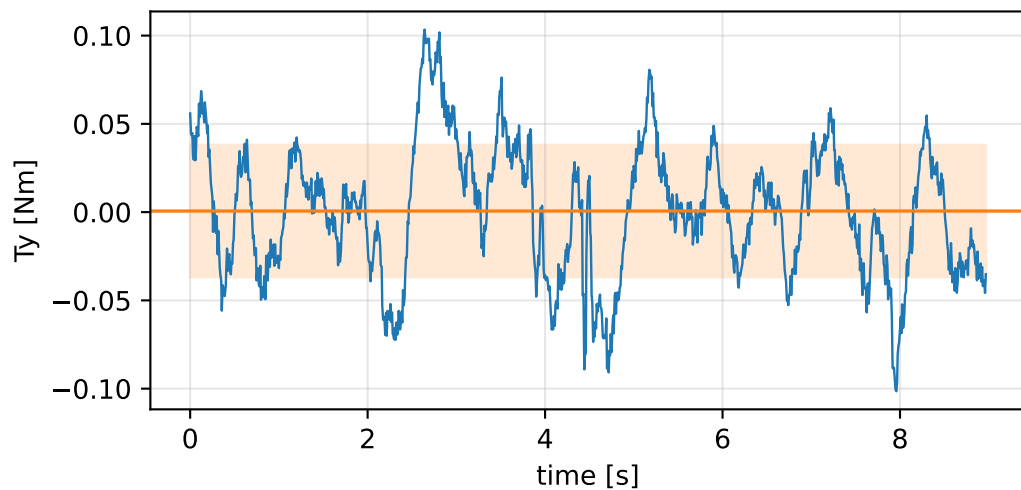
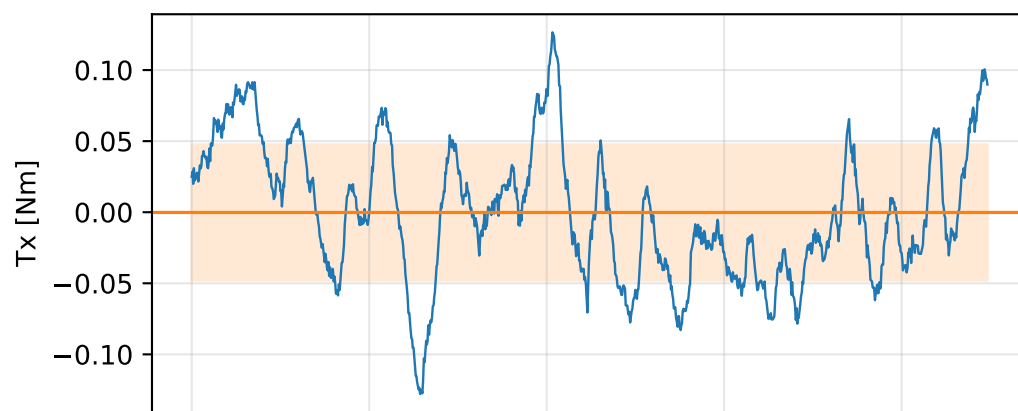
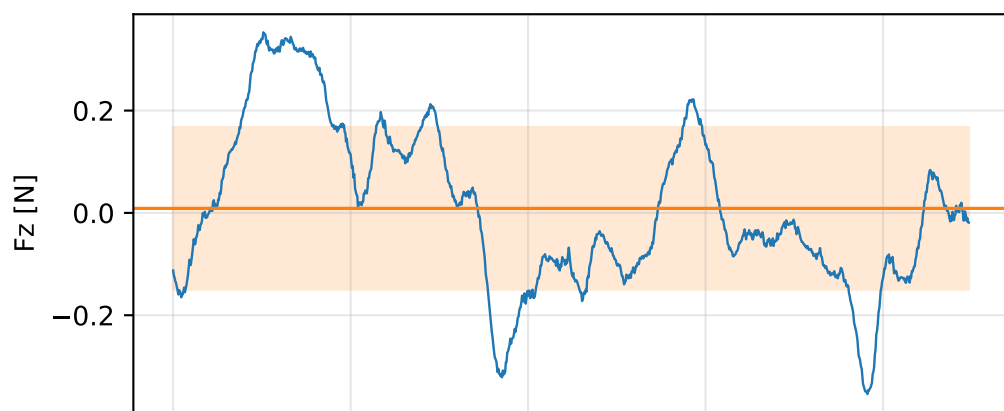
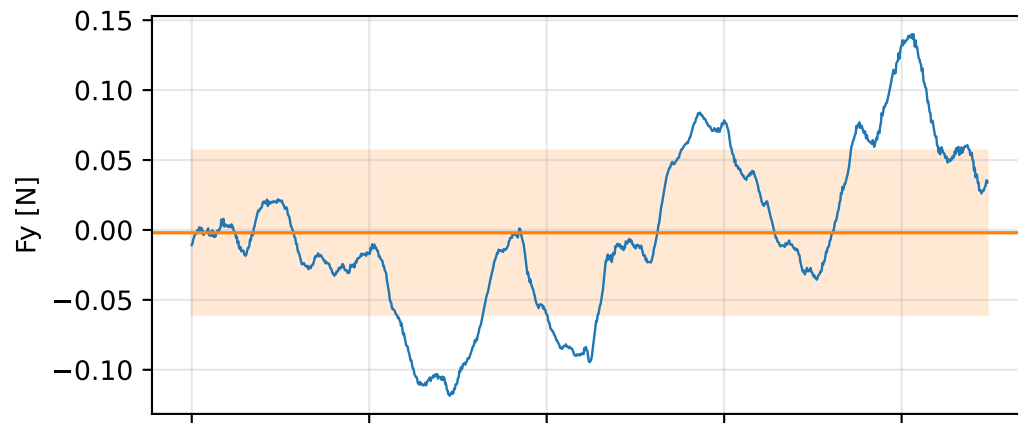
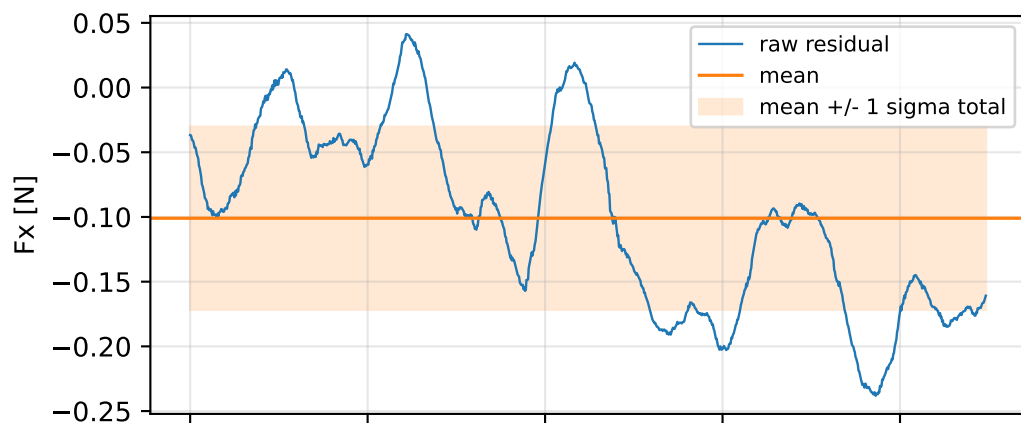


inertia_over_mass_xz_nominal: SG trajectory and derived motion



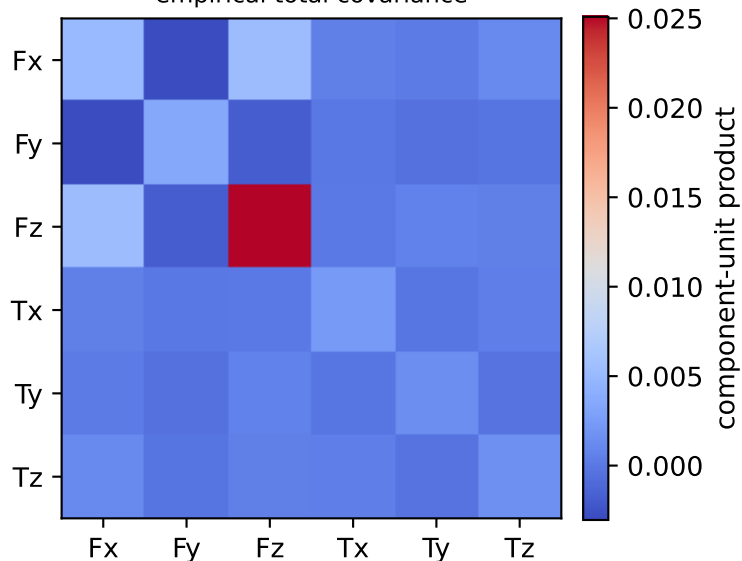
inertia_over_mass_xz_nominal: raw Newton--Euler residual wrench (not trajectory-fitted external wrench)
mass gauge fixed to nominal mass = 2.35159759 kg



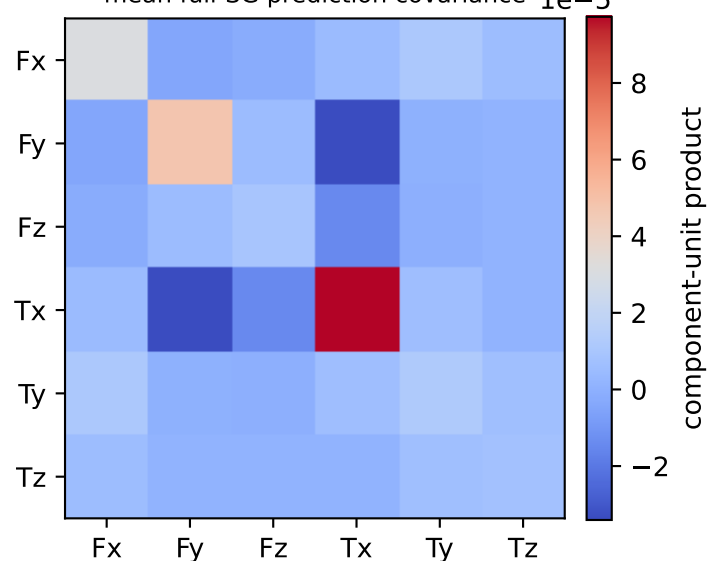
vector RMS about zero: force 0.20917 N, torque 0.07226 Nm; component std about mean: force [0.0706 0.0586 0.1584], torque [0.0477 0.0374 0.0394]

inertia_over_mass_xz_nominal: residual-wrench covariance decomposition in nominal-mass gauge
force [N], torque [Nm]; raw excess eigenvalues [0.001 0.001 0.002 0.003 0.006 0.027]

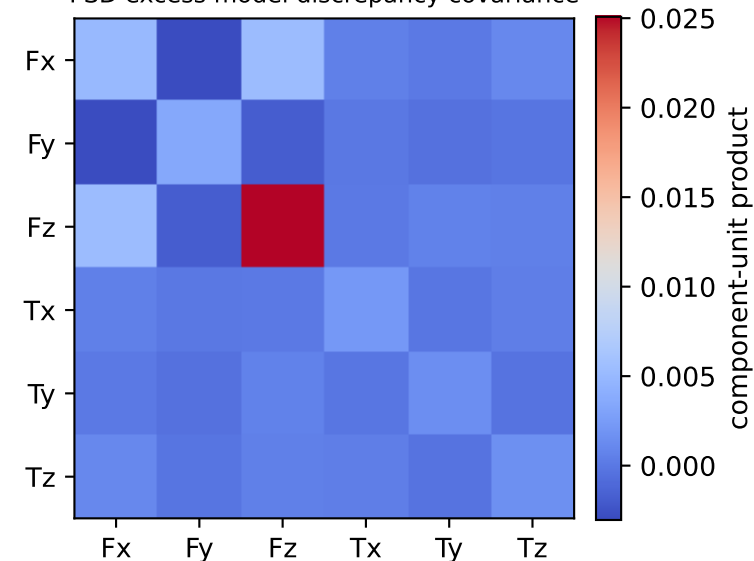
empirical total covariance



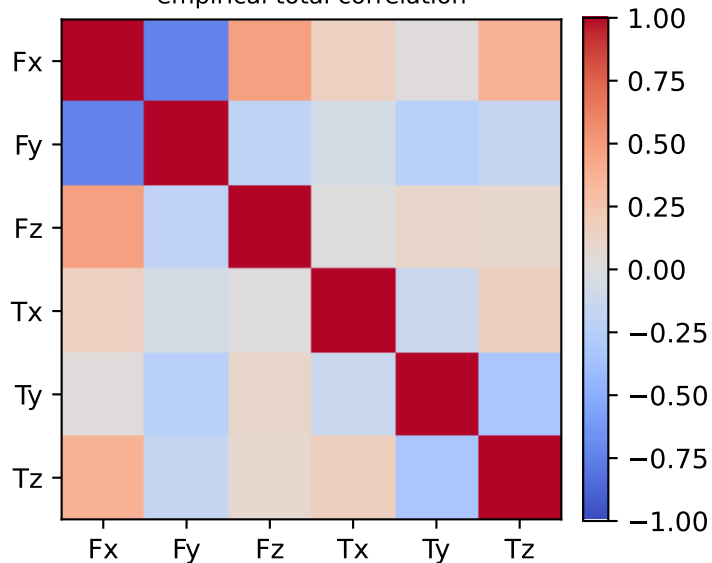
mean full-SG prediction covariance $1e-5$



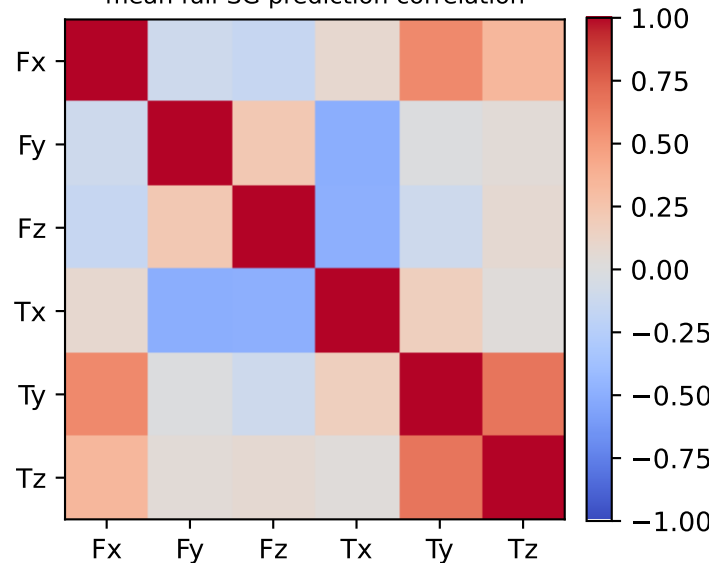
PSD excess model discrepancy covariance



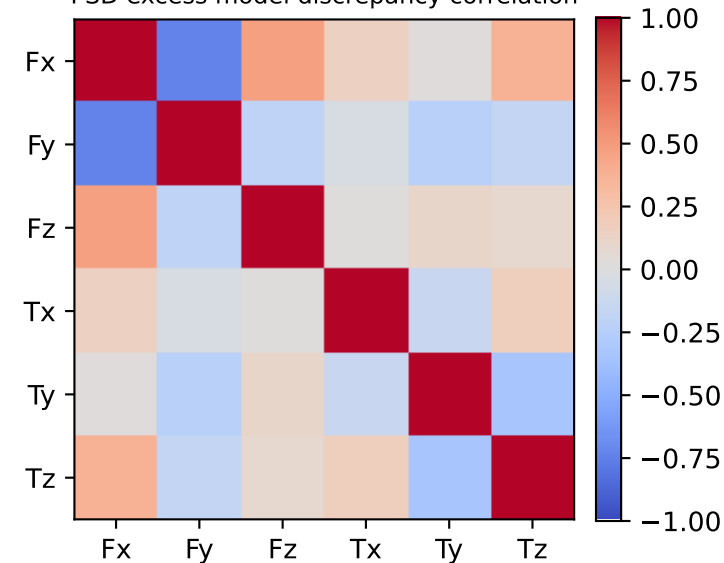
empirical total correlation



mean full-SG prediction correlation

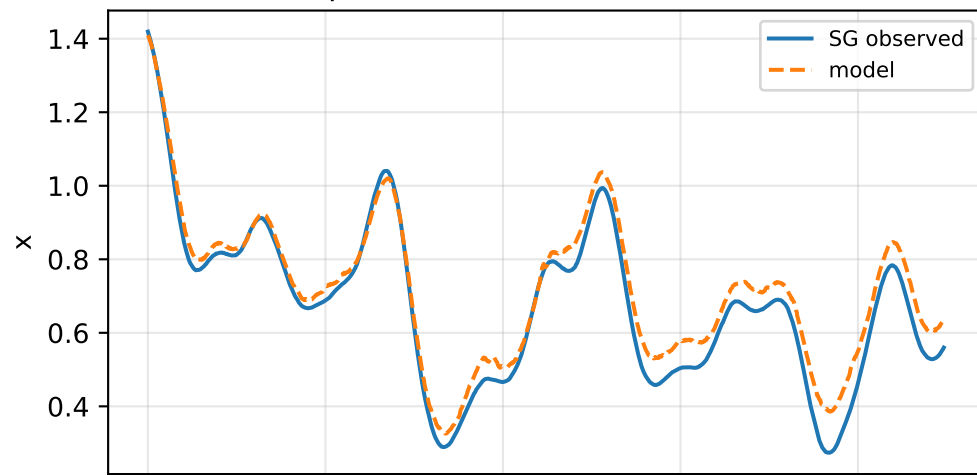


PSD excess model discrepancy correlation

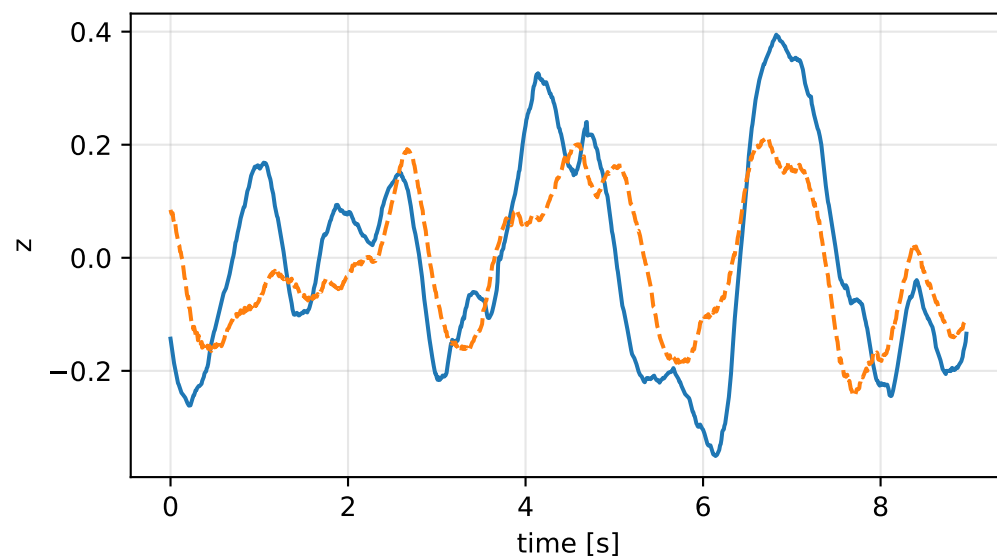
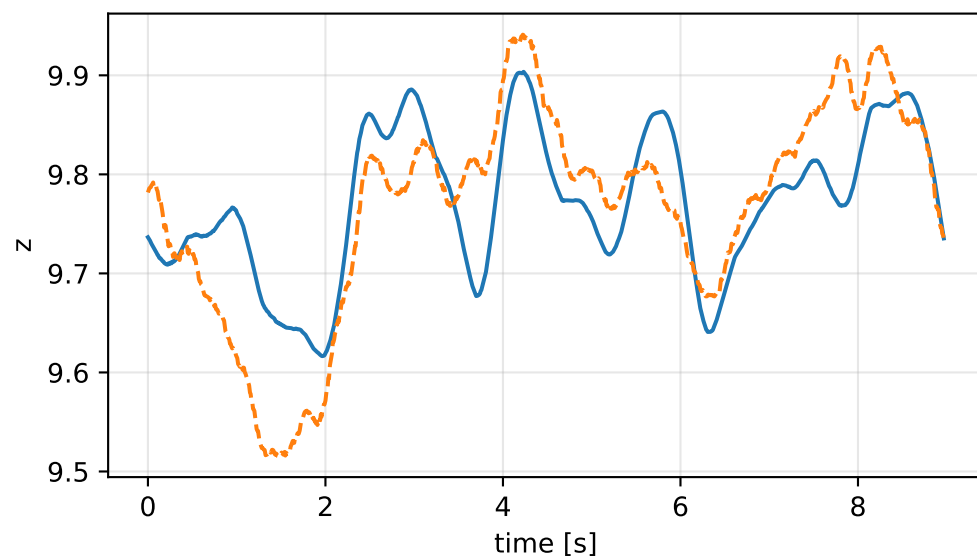
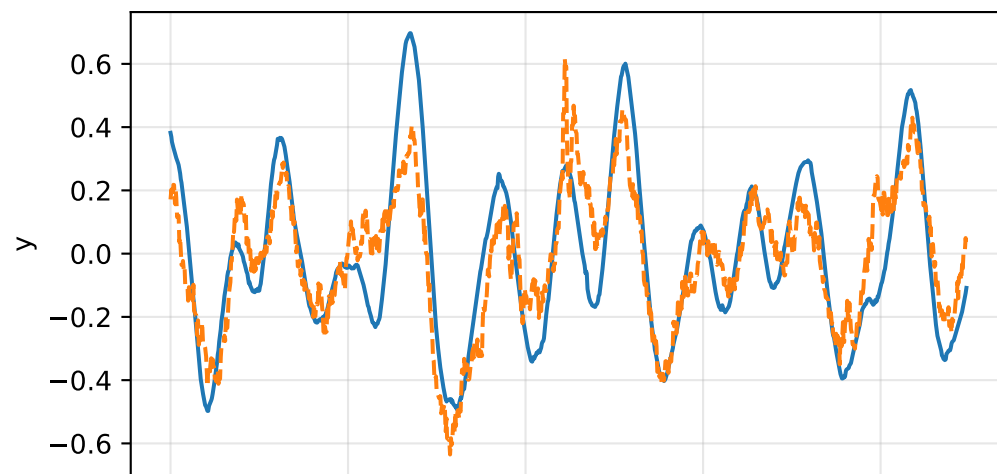
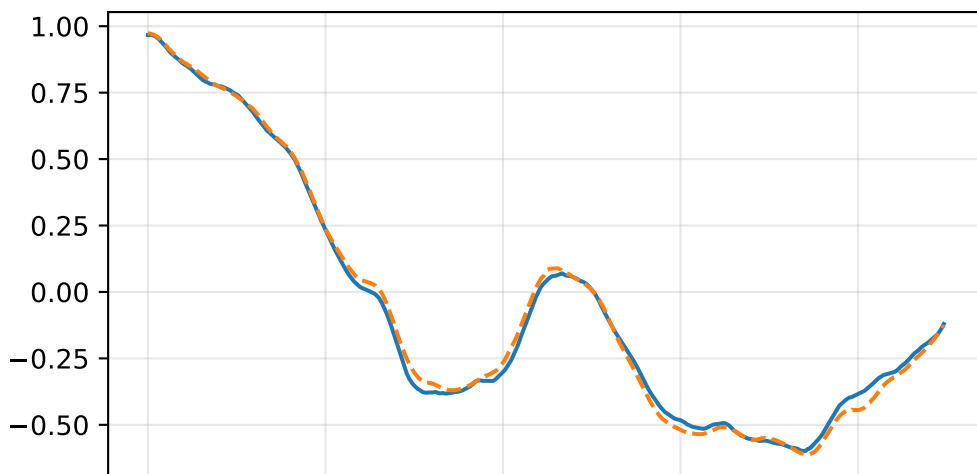
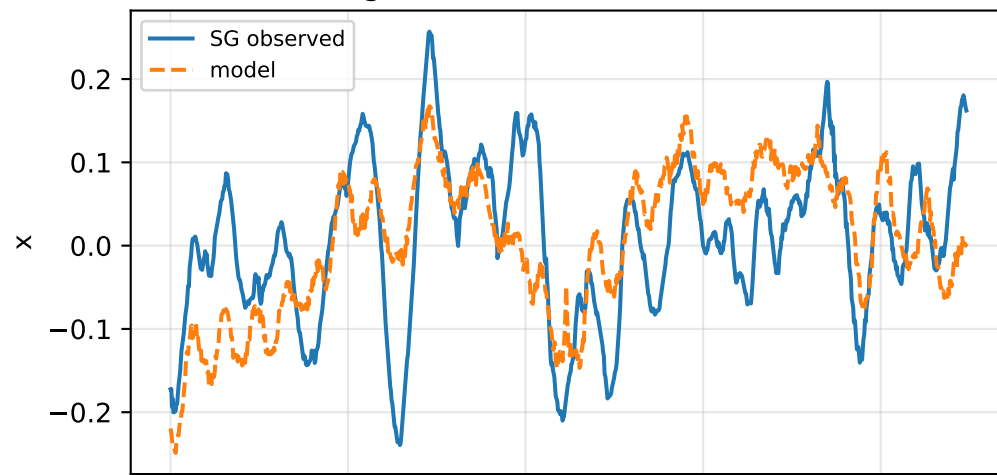


inertia_over_mass_xz_nominal: acceleration residual objective

specific acceleration [m/s²]

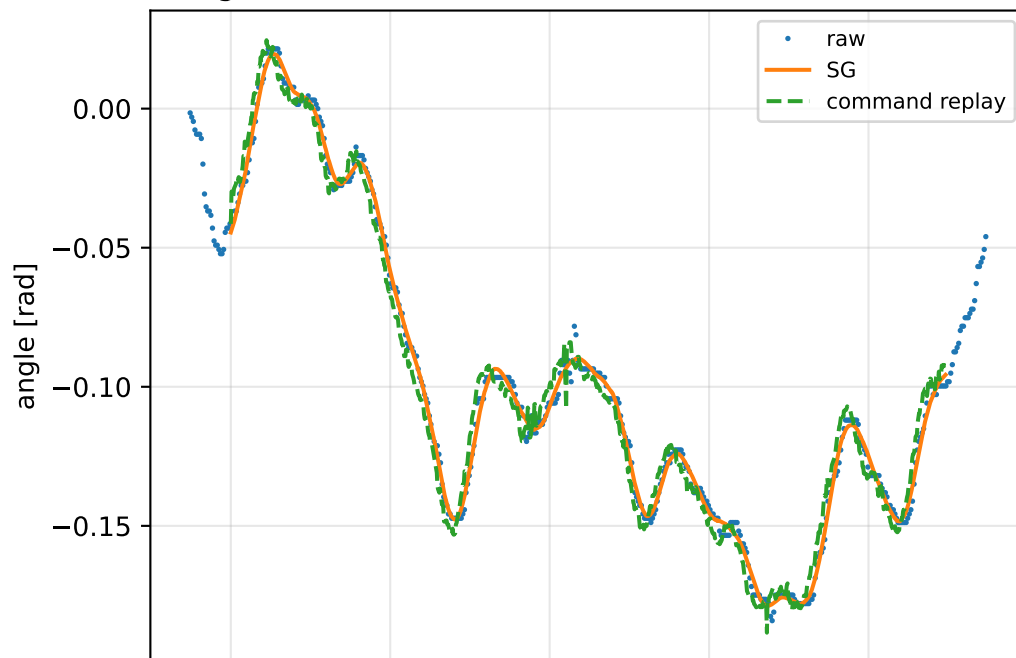


angular acceleration [rad/s²]

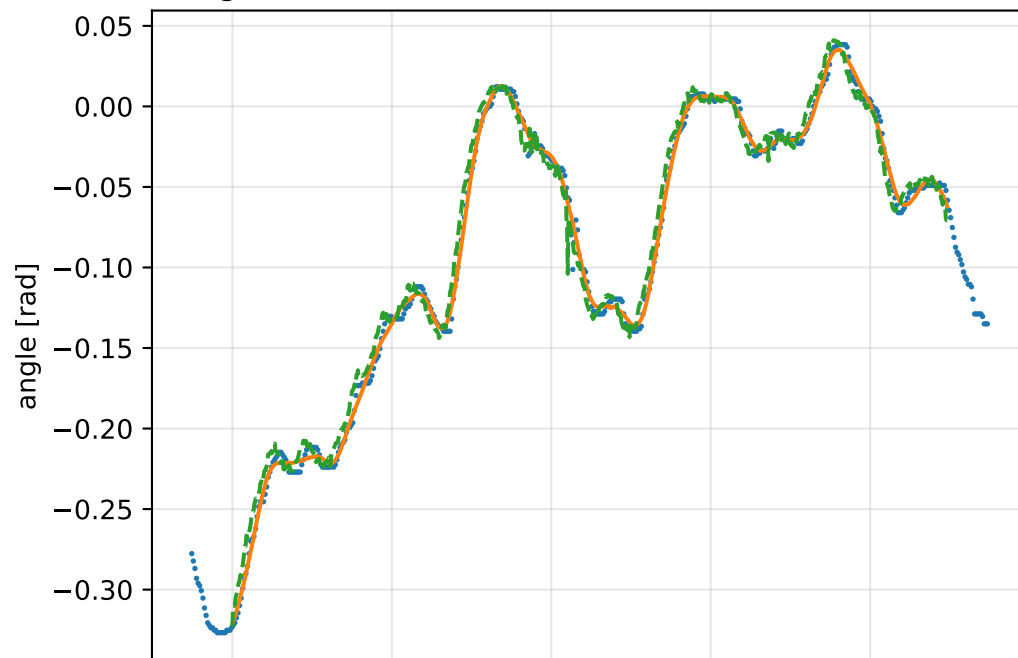


inertia_over_mass_xz_nominal: gimbal measurement audit (objective=measured_sg)

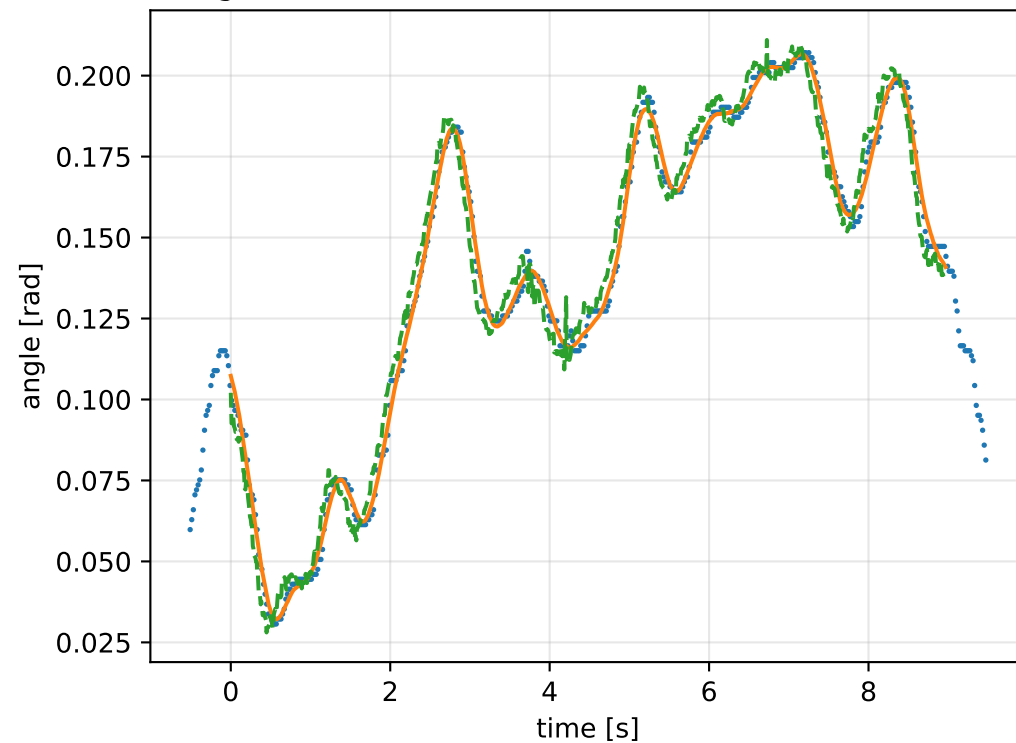
gimbal 1: RMSE=0.006345, max=0.01494 rad



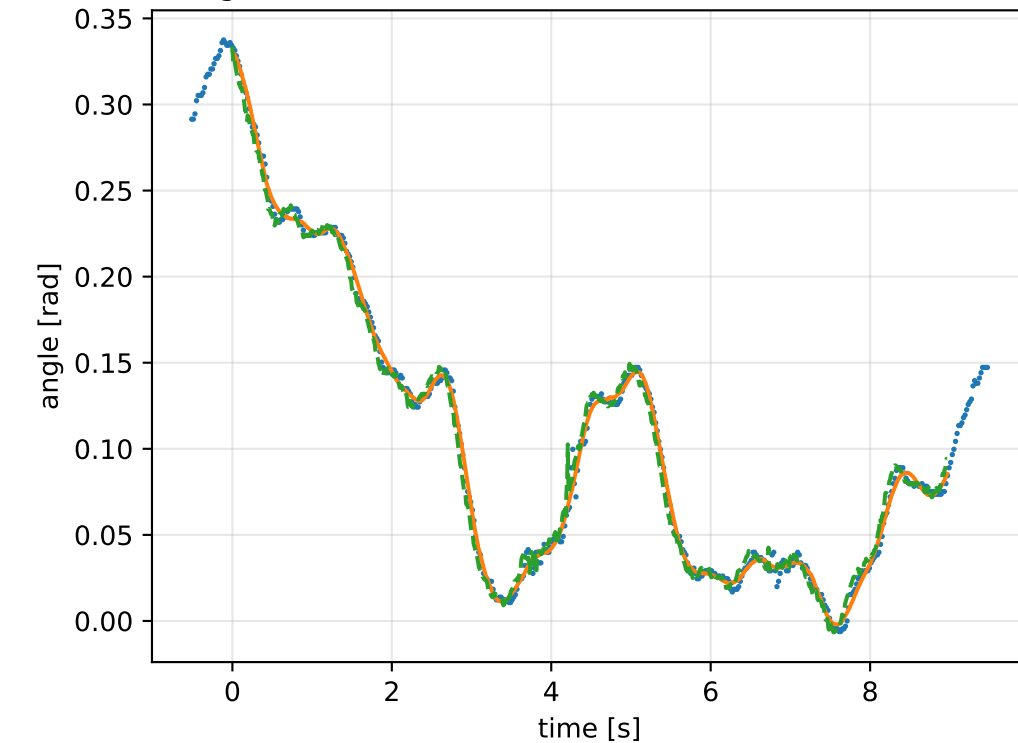
gimbal 2: RMSE=0.008835, max=0.04462 rad



gimbal 3: RMSE=0.007019, max=0.01611 rad

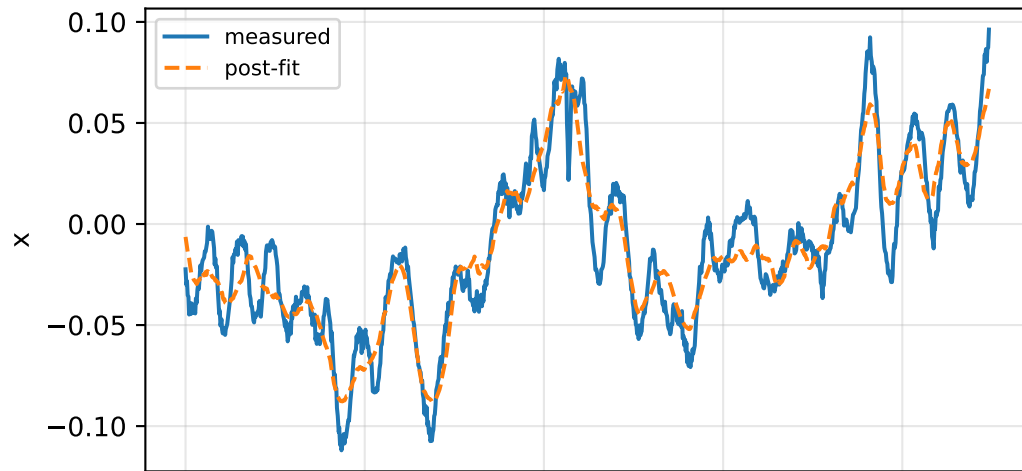


gimbal 4: RMSE=0.006321, max=0.03642 rad

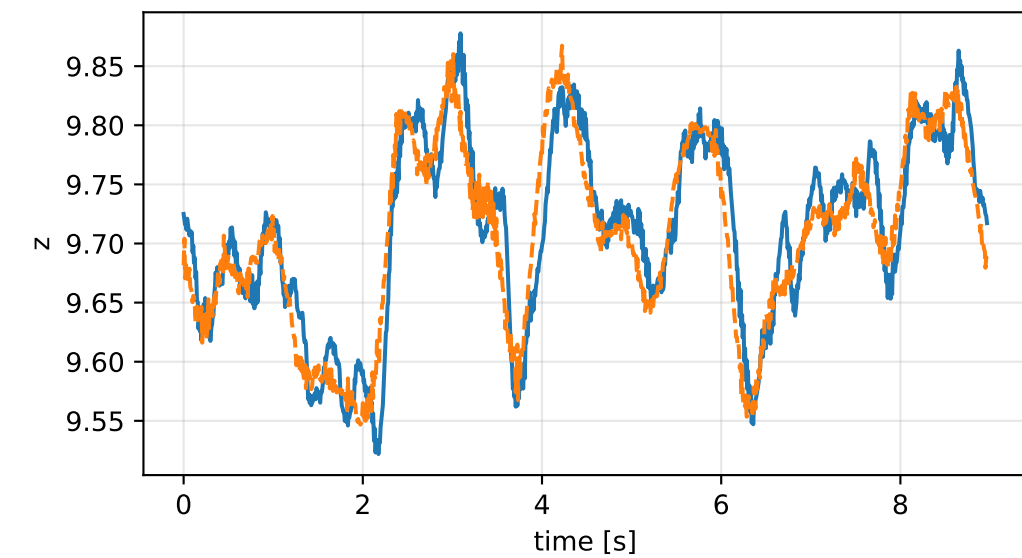
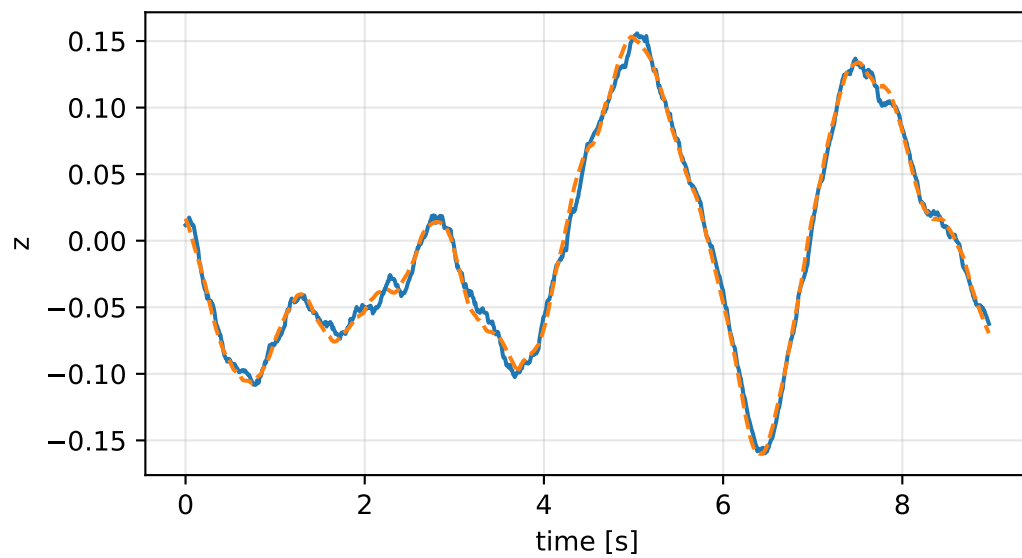
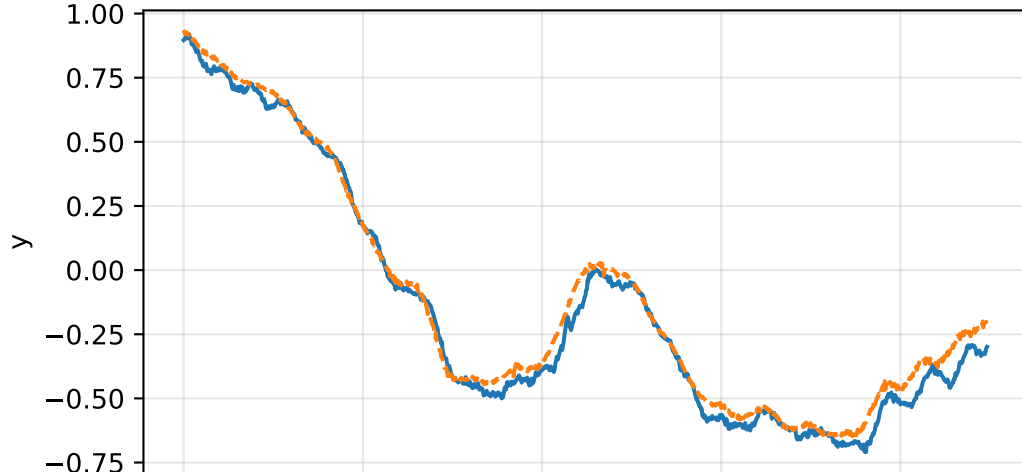
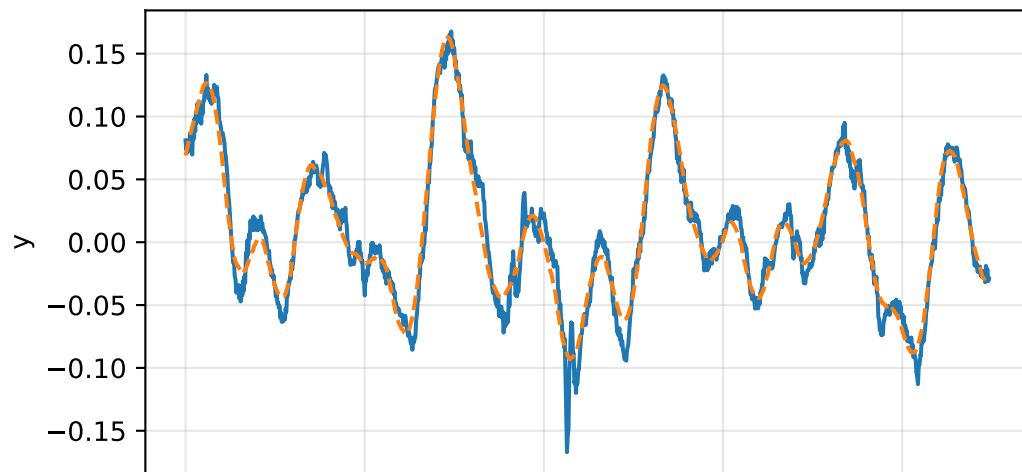
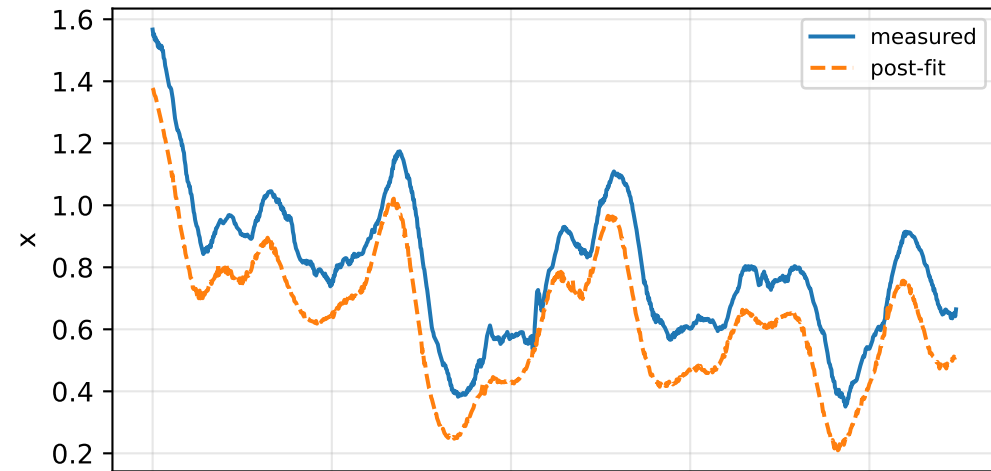


inertia_over_mass_xz_nominal: IMU comparison (diagnostic only)

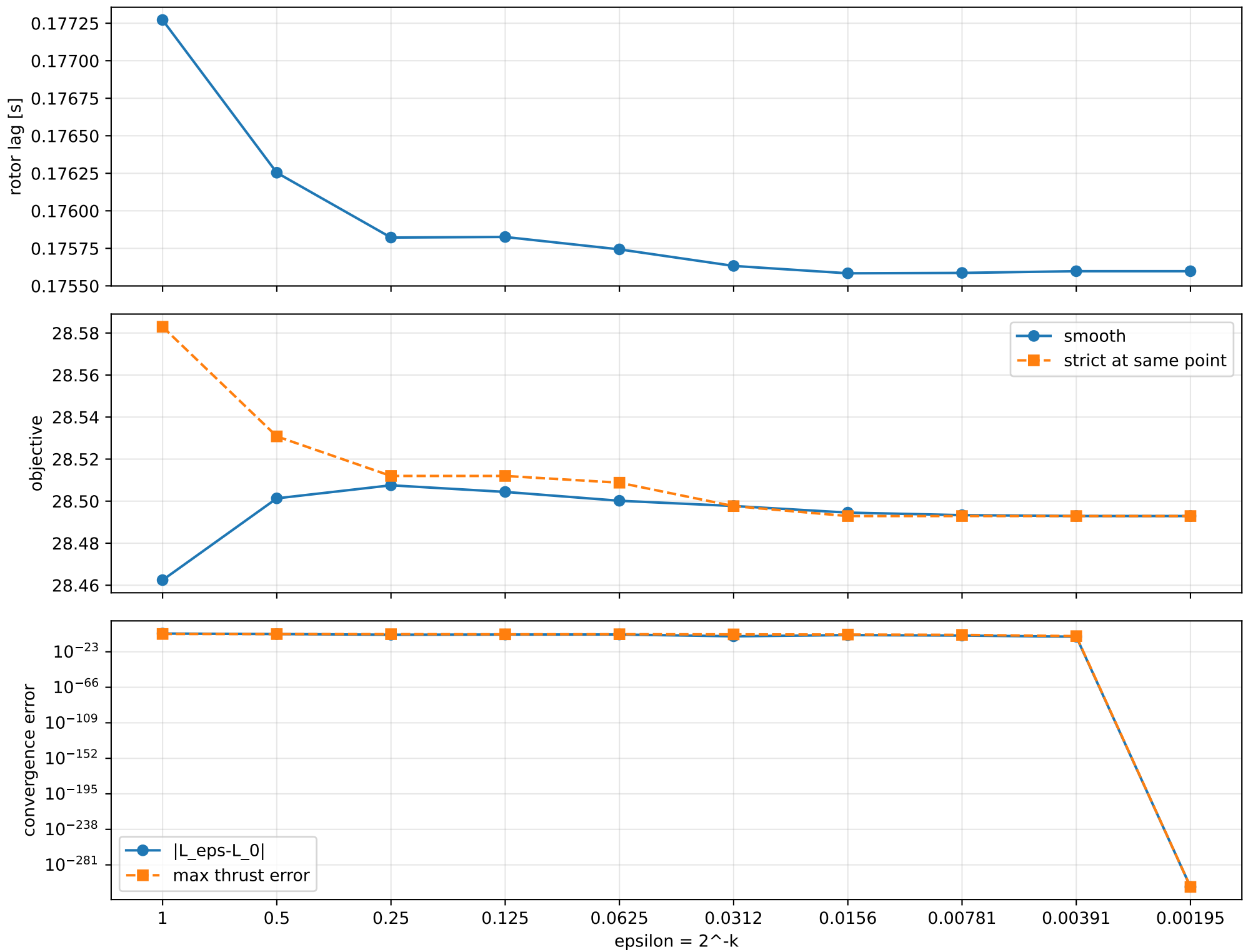
gyro [rad/s]



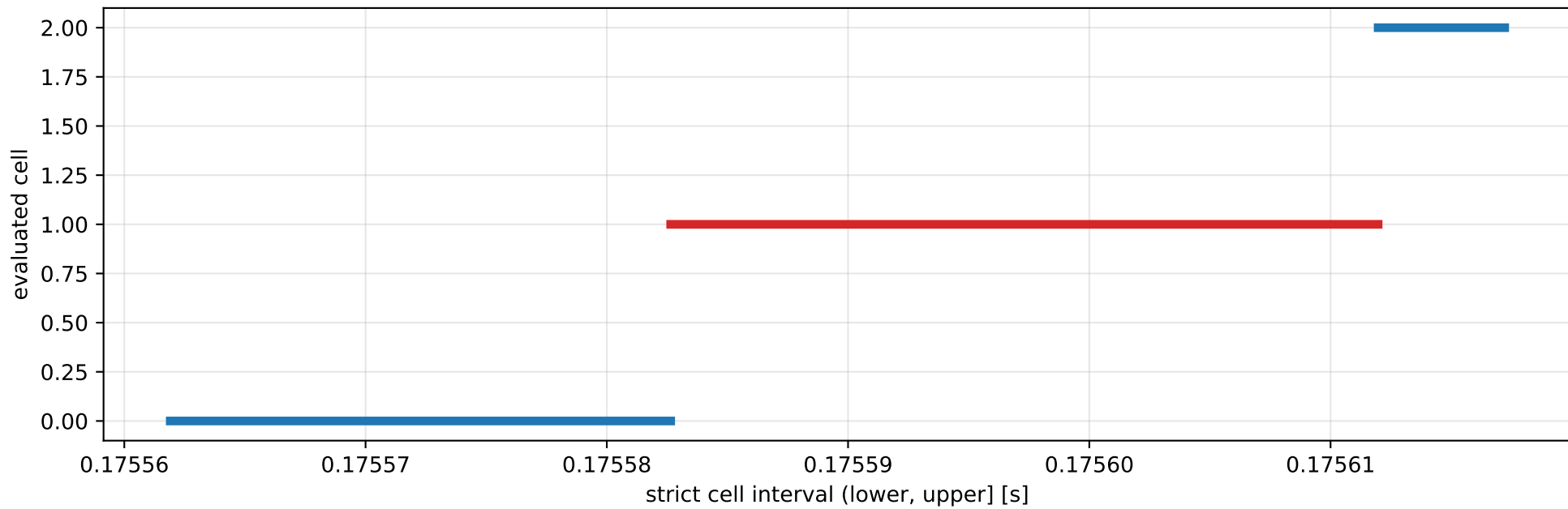
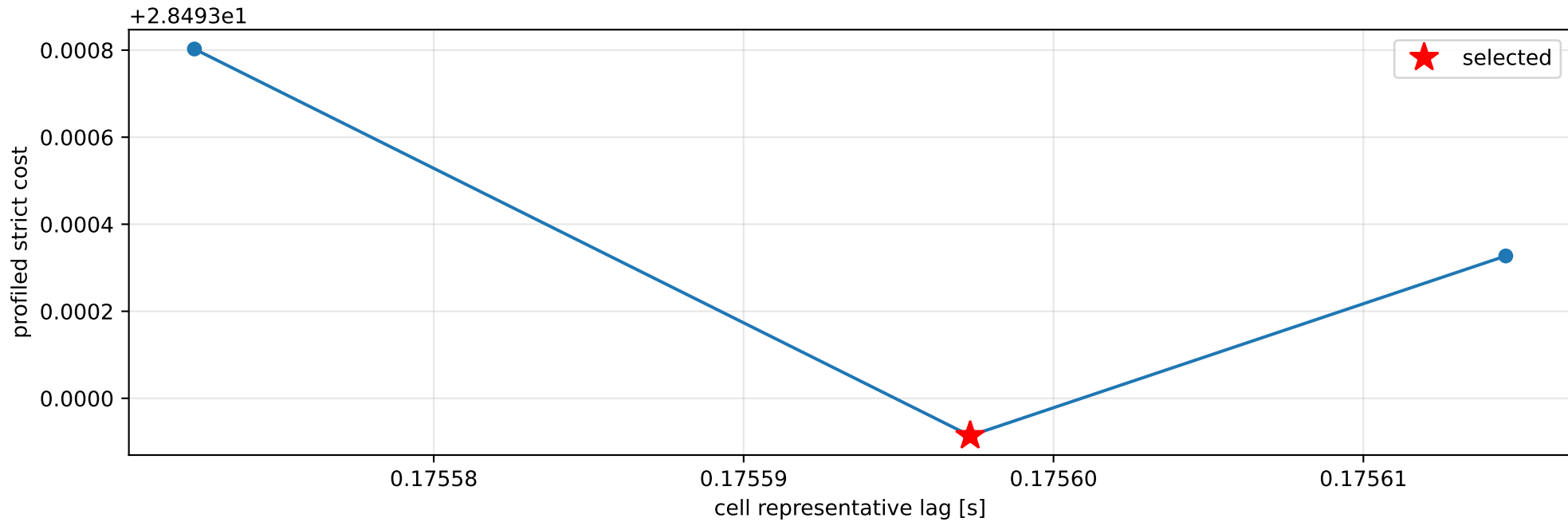
specific force [m/s^2]



inertia_over_mass_xz_nominal: smooth-to-strict continuation



inertia_over_mass_xz_nominal: exact strict-ZOH lag cells



selected (0.175582647, 0.175611973] s; final smooth 0.175597721 s; data boundary=False

inertia_over_mass_xz_nominal: parameter estimate

Case: inertia_over_mass_xz_nominal

status: completed (all stages completed)

mass [kg]: nominal 2.35159759; estimated 5.42518186

inertia [kg m²]:

[[1.28529505e+00 -1.34726740e-02 4.38682493e-05]

[-1.34726740e-02 5.96913287e-01 -1.72089387e-02]

[4.38682493e-05 -1.72089387e-02 6.88909805e-01]]

CoG body [m]: [-0.01654605 -0.05566426 -0.00488202]

force effectiveness: [2.38878488 2.39384904 1.26933402 1.48244501]

scale-free J/m [m²]:

[[2.36912806e-01 -2.48335896e-03 8.08604218e-06]

[-2.48335896e-03 1.10026411e-01 -3.17204826e-03]

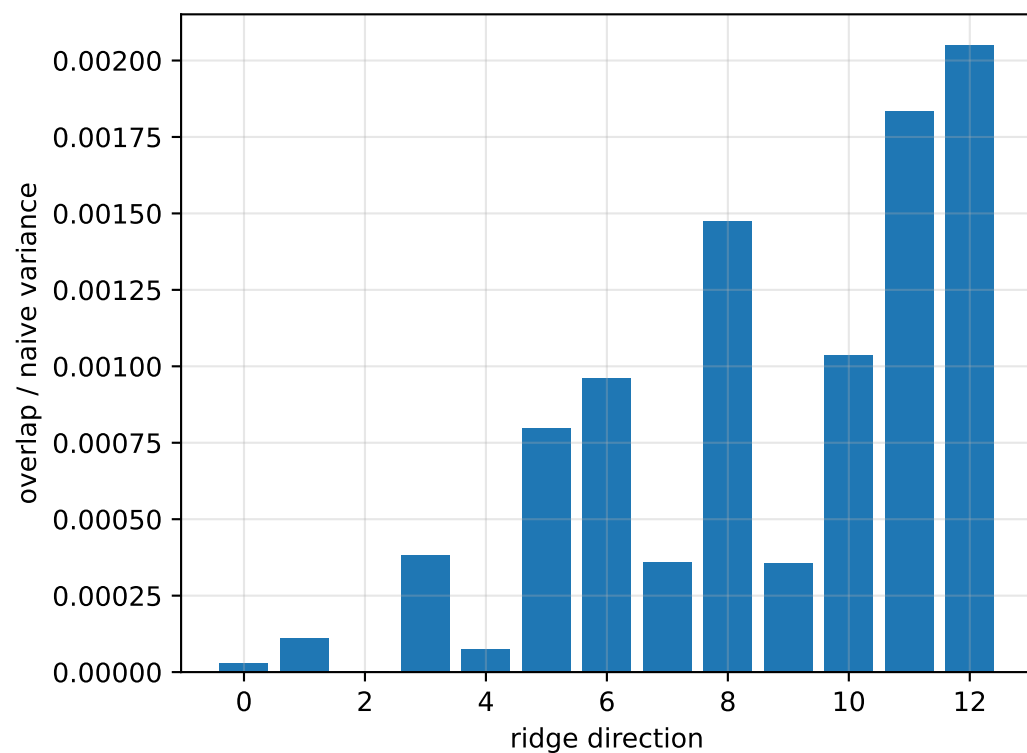
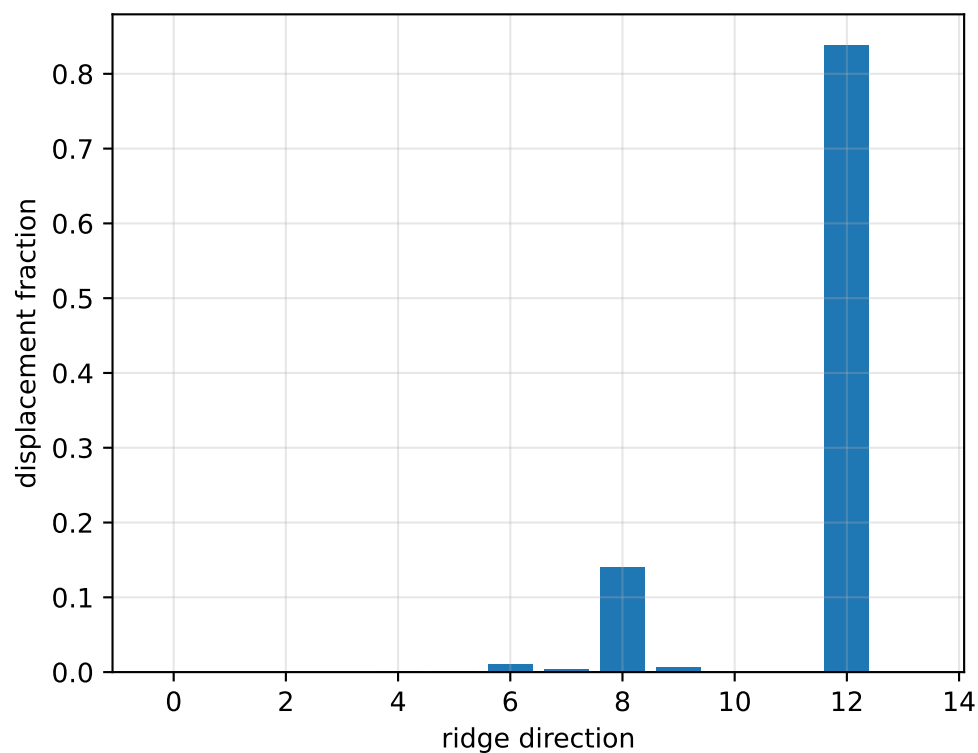
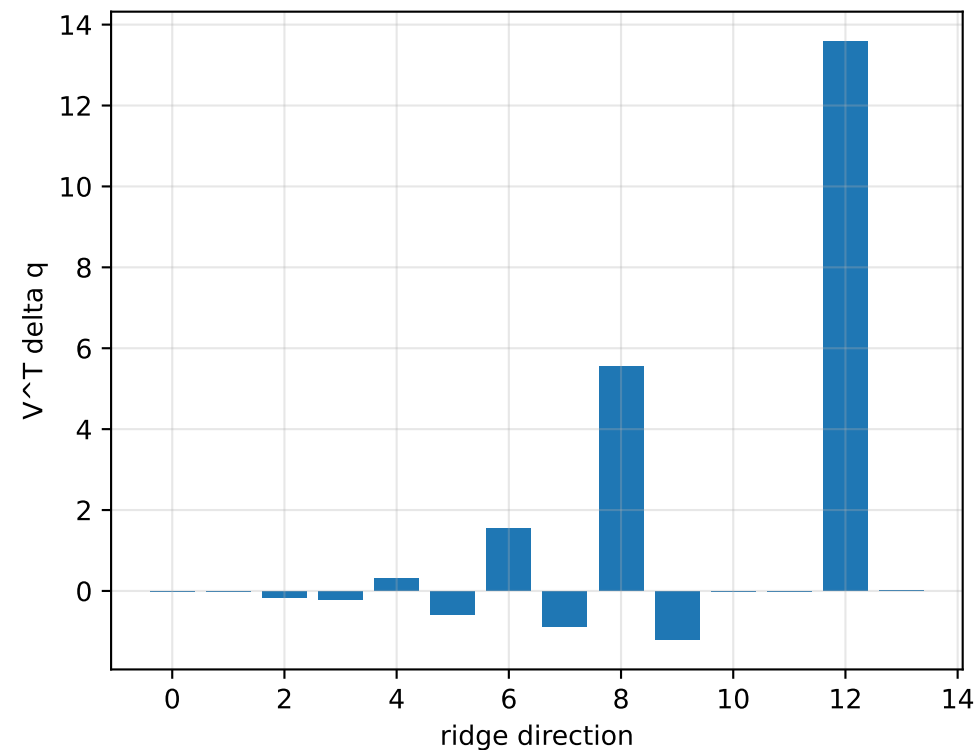
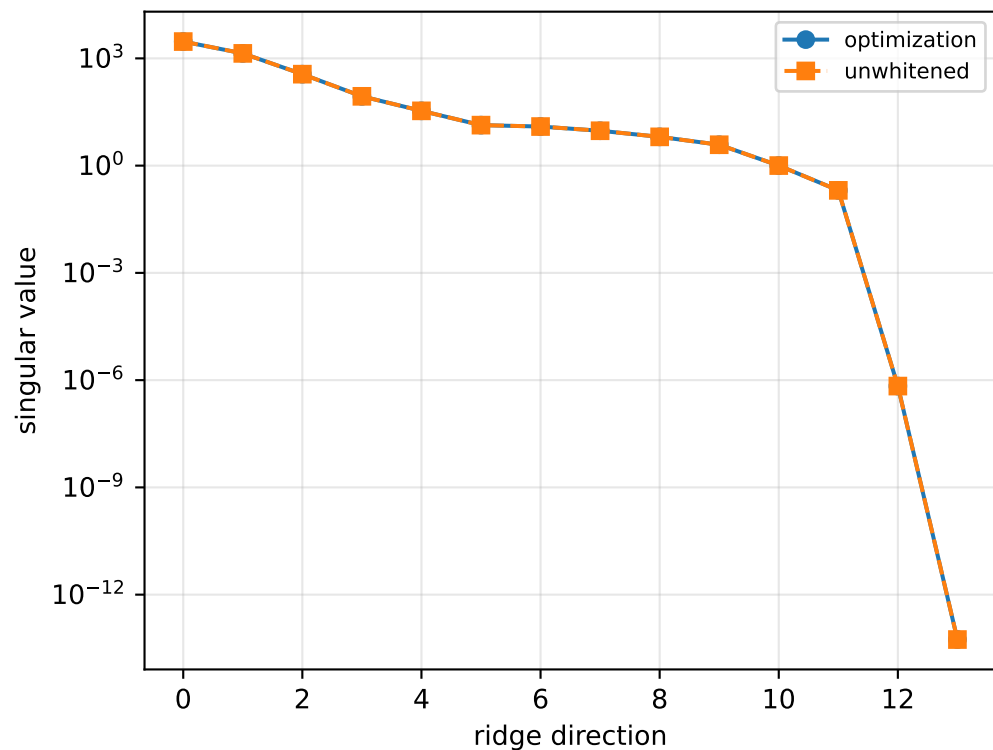
[8.08604218e-06 -3.17204826e-03 1.26983726e-01]]

scale-free f/m: [0.44031425 0.4412477 0.23397078 0.27325259]

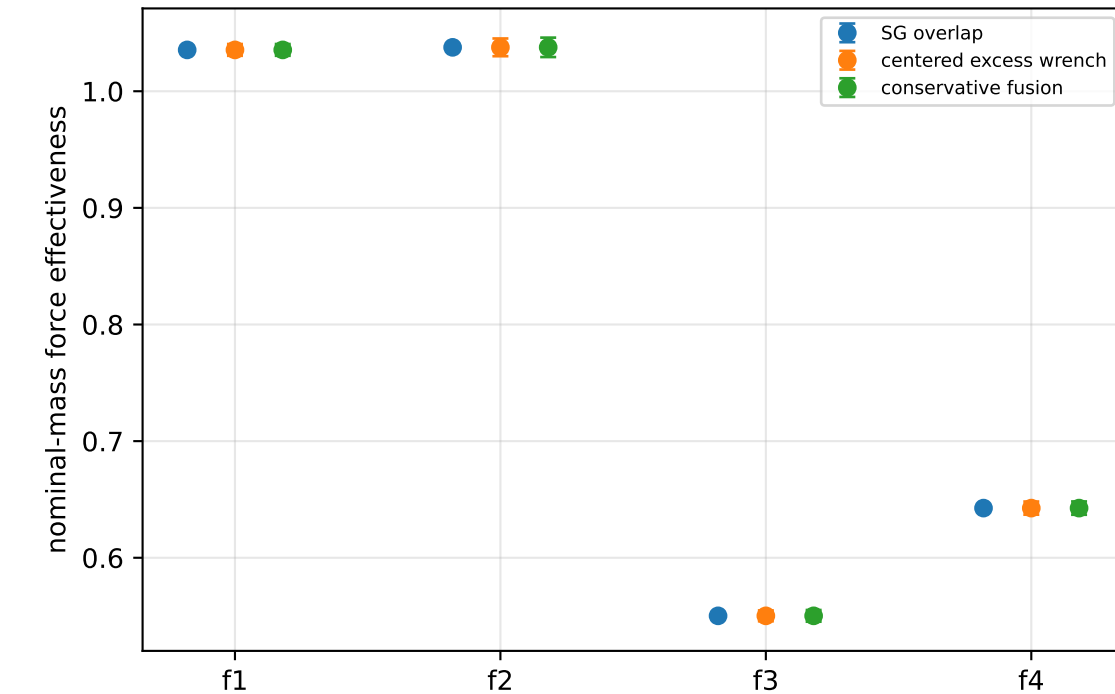
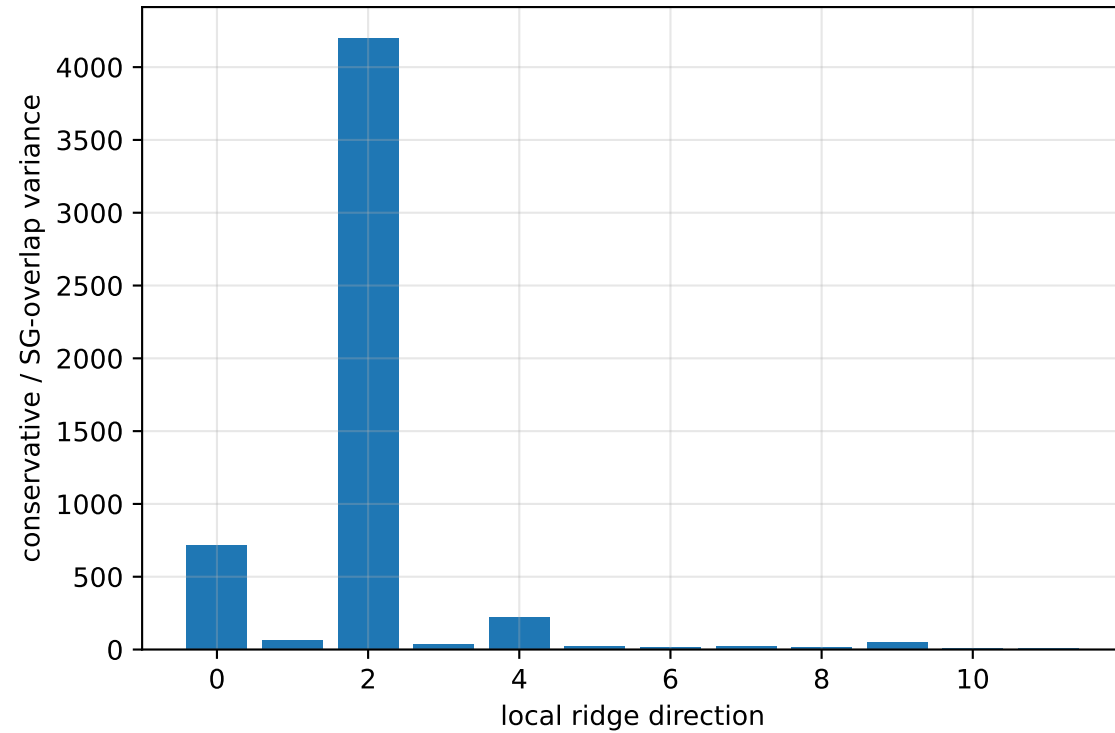
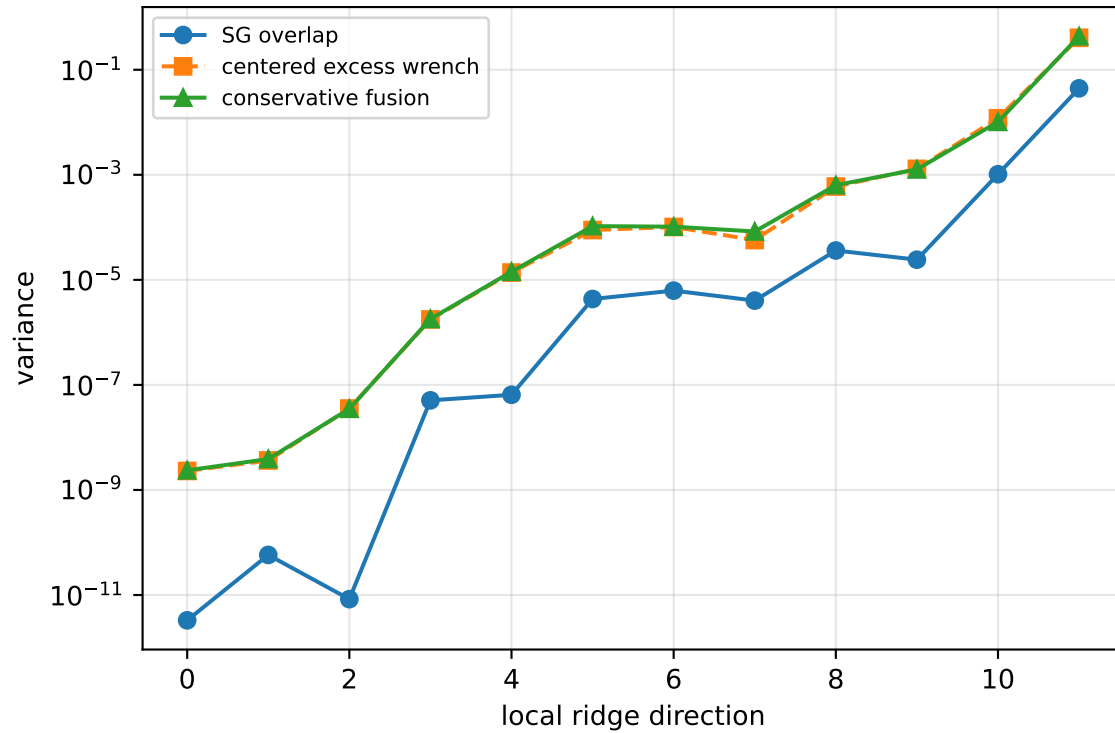
rotor lag cell: (0.175582647, 0.175611973] s

representative: 0.17559731 s

inertia_over_mass_xz_nominal: ridge spectrum (rank 13, nullity 1, || $\mathbf{v}$ ||=7.64e-14)



inertia_over_mass_xz_nominal: Conservative fusion uncertainty



Top three non-gauge ambiguous directions
 $\text{tr}(\mathbf{M}_{\text{res}}) / \text{tr}(\mathbf{M}_{\text{SG}}) = 418.7$
wrench-to-acceleration recovery max error = $2.8\text{e-}15$
exact scale gauge ridge index = 13 (alignment 1)

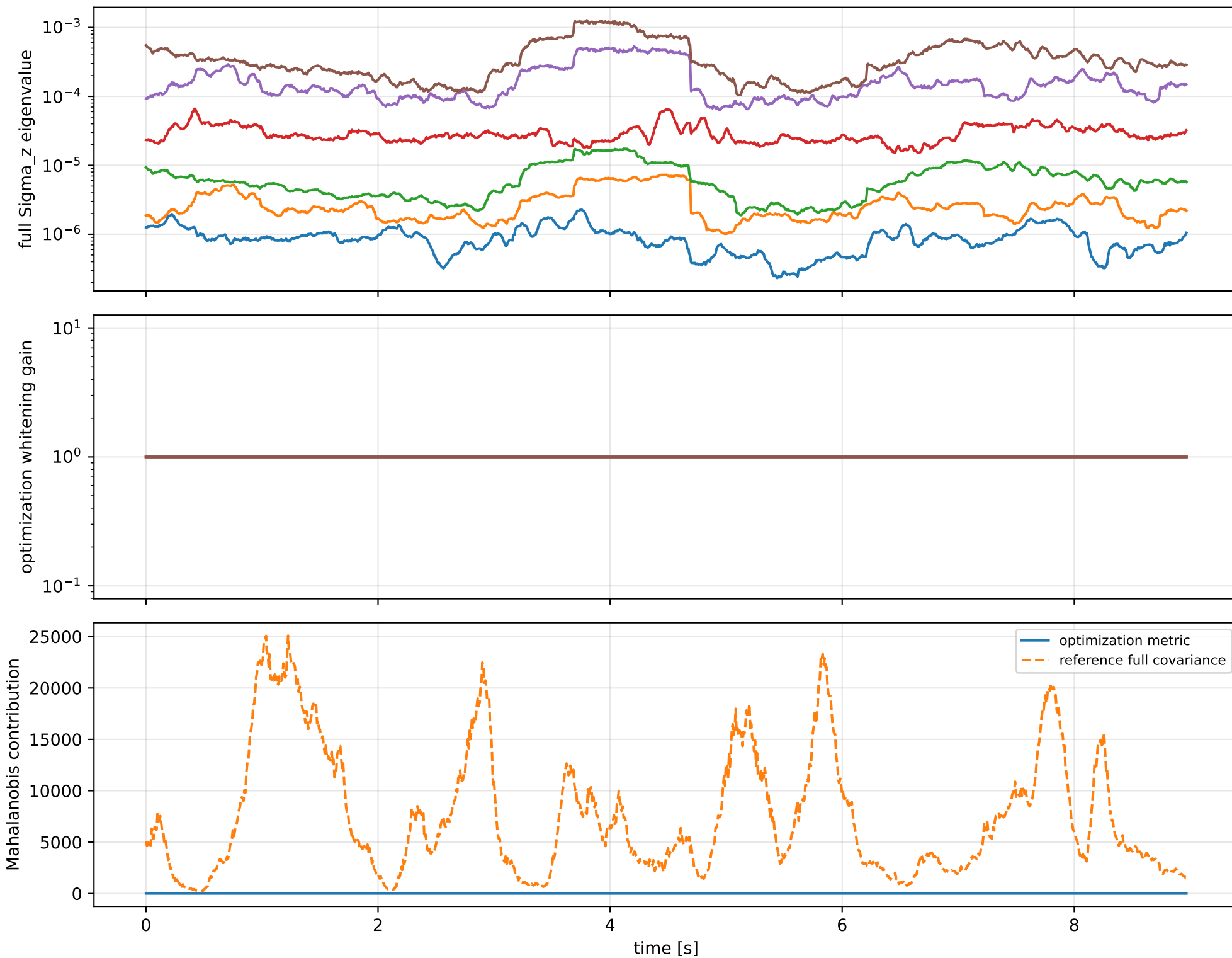
index 11: variance=0.4338, ratio=9.803
 $\log_{\text{mass}}=+0.000508$; second_moment_diag_1=+0.0275
second_moment_diag_2=-0.0281; second_moment_diag_3=-0.00244
second_moment_offdiag_12=+0.994; second_moment_offdiag_13=-0.0988
second_moment_offdiag_23=+0.00012; cog_x=-0.0006
cog_y=+0.000256; cog_z=-0.000557
log_force_eff_1=+0.00244; log_force_eff_2=-0.0035
log_force_eff_3=+0.000158; log_force_eff_4=+0.00342

index 10: variance=0.01017, ratio=9.92
 $\log_{\text{mass}}=-6.75\text{e-}05$; second_moment_diag_1=-0.00303
second_moment_diag_2=+0.0188; second_moment_diag_3=-0.00646
second_moment_offdiag_12=-0.0983; second_moment_offdiag_13=-0.993
second_moment_offdiag_23=+0.0121; cog_x=-0.00651
cog_y=+0.000771; cog_z=-0.000646
log_force_eff_1=+0.0193; log_force_eff_2=-0.0266
log_force_eff_3=-0.0407; log_force_eff_4=+0.0388

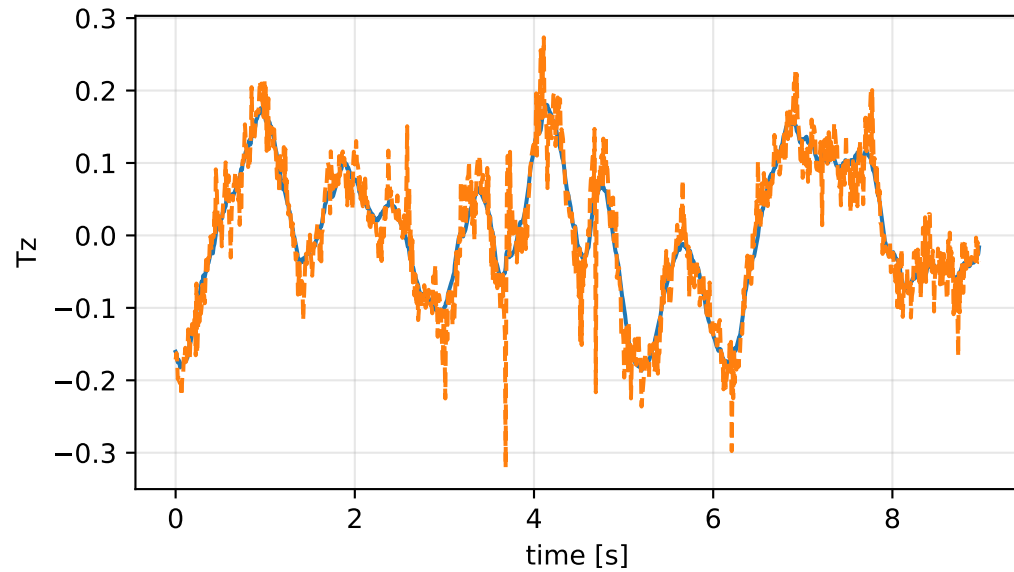
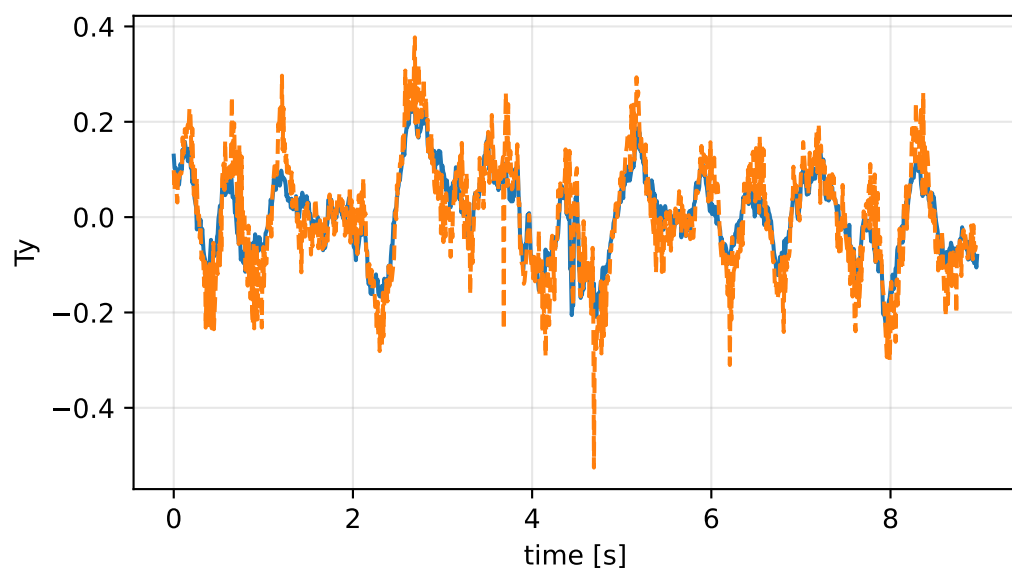
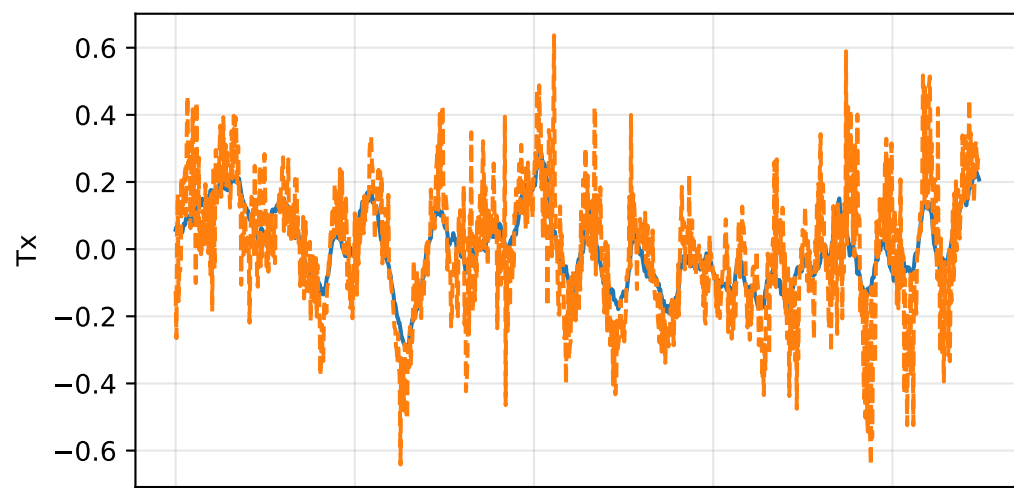
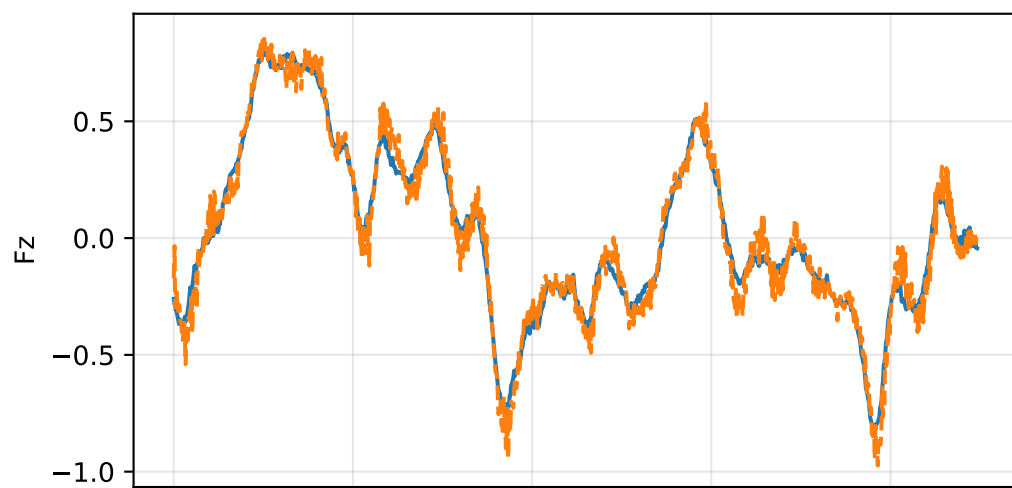
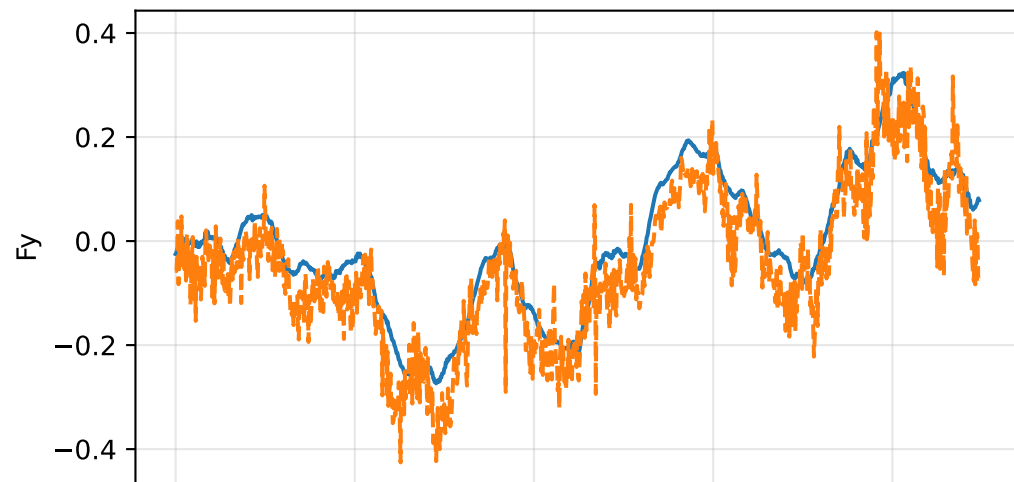
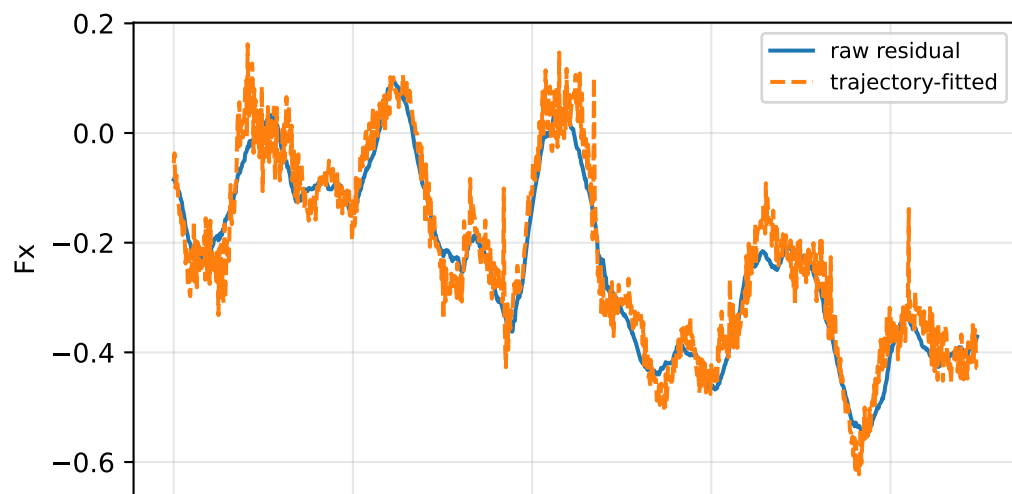
index 9: variance=0.001266, ratio=52.3
 $\log_{\text{mass}}=+0.166$; second_moment_diag_1=-0.00043
second_moment_diag_2=-0.899; second_moment_diag_3=+0.0561
second_moment_offdiag_12=-0.0278; second_moment_offdiag_13=-0.0221
second_moment_offdiag_23=-0.206; cog_x=+0.0047
cog_y=+0.000276; cog_z=+0.0109
log_force_eff_1=+0.16; log_force_eff_2=+0.172
log_force_eff_3=+0.21; log_force_eff_4=+0.135

The conservative fusion covariance deliberately retains the existing SG-overlap uncertainty and adds the uncentered empirical residual score second moment without subtracting the SG contribution. It is intended as a conservative fusion distribution, not as a calibrated generative noise covariance.

inertia_over_mass_xz_nominal: optimization vs reference covariance



inertia_over_mass_xz_nominal: wrench / closure (force max $1.52\text{e-}14$, torque max $1.04\text{e-}16$)



inertia_over_mass_xz_nominal: optional parameter-prior audit

Optional physical parameter prior

The parameter prior is optional external information. The default estimator is prior-free.

name: pseudo_condition_inertia_over_mass_xz_nominal

role: pseudo_conditioning_ablation

source: /home/leus/catkin_ws/src/ProbTF-demo/ros/examples/grape-param-estim/minimal/config/priors/pseudo_conditioning/scalar/inertia_over_ma

SHA256: 99d969ed0e0fa85ae900b089864493fcb7e345fdd5fd63009a6ebdbc6ee04171

data objective: 28.4929140472

prior objective: 1.25633296581e-10

total objective: 28.4929140473

This configuration uses an intentionally tight artificial standard deviation for an ablation-style pseudo-conditioning experiment; it is not a calibrated physical uncertainty.

factor: inertia_over_mass_xz_nominal (inertia_over_mass_m2)

components: xz

target: [8.08602633e-06]

std: [1.e-06]

final: [8.08604218e-06]

error: [1.58513909e-11]

standardized residual: [1.58513909e-05]

factor objective: 1.25633296581e-10